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An investigation of the imputation techniques for missing values in ordinal data enhancing clustering and classification analysis validity

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ABSTRACT

Missing data can significantly impact dataset integrity and suitability, leading to unreliable statistical results, distortions, and poor decisions. The presence of missing values in data introduces inaccuracies in clustering and classification and compromises the reliability and validity of such analyses. This study investigates multiple imputation techniques specifically designed for handling missing values in ordinal data commonly encountered in surveys and questionnaires. Quantitative approaches are used to evaluate different imputation methods on various datasets with varying missing value percentages. By comparing the performance of imputation techniques using clustering metrics and algorithms (e.g., k-means, Partitioning Around Medoids), the study provides valuable insights for selecting appropriate imputation methods for accurate data analysis. Furthermore, the study examines the impact of imputed values on classification algorithms, including k-Nearest Neighbors (kNN), Naive Bayes (NB), and Multilayer Perceptron (MLP). Results demonstrate that the decision tree method is the most effective approach, closely aligning with the original data and achieving high accuracy. In contrast, random number imputation performs poorly, indicating limited reliability. This study advances the understanding of handling missing values and emphasizes the need to address this issue to enhance data analysis integrity and validity.

1. Introduction

Missing data can severely hinder effective decision-making [1]. Particularly within the realm of large, real-world datasets, the absence of data is a common and inherent challenge. Such gaps in data prevent comprehensive analysis and obstruct the generation of valuable inferences necessary for decision-making [2]. Existing data analysis methodologies may fall short or produce inaccurate results when faced with missing values. Consequently, it is critical to develop strategies to manage these missing values effectively, thereby enhancing decision-making systems [1]. In instances like surveys and questionnaires, multiple factors contribute to missing values, ranging from poorly designed surveys and deliberate non-responsiveness from participants to the omission of sensitive data for confidentiality purposes [3].

Recent advancements in data analysis techniques offer a myriad of methods to tackle the issue of missing data. One such simplistic approach is to discard data vectors containing missing values. However, this technique, despite its simplicity, can introduce biases in the resulting analysis and is impractical for datasets with substantial missing data [4]. In the real world, datasets often contain missing values that obstruct statistical inferences and decision-making [3]. It is thus critical to assign values – whether arbitrary or informed – to complete these datasets [5]. Imputation is a solution to missing values, wherein missing data is replaced by plausible approximations [6]. Uninformed patterns of absent data commonly appear in real-world datasets [7]. The chosen imputation technique should aim to improve data analysis and deliver unbiased results [8]. Simple techniques such as replacing missing values with the dataset's mean, median, or mode exist, but there are also more complex classification-based methods [8].

Ordinal data refers to observations that consist of variables ranked on a scale [5]. Examples include scales such as high, medium, and low. While ordinal values can be ordered, their distances cannot be measured. On the other hand, missing values refer to the absence of data due to various factors such as transmission errors, non-responses, data entry errors, or data capture errors [9]. These missing values can

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have a significant impact on the insights that can be derived from the data.

Missing data can be categorized into three types: Missing Completely at Random (MCAR), Missing at Random (MAR), and Missing Not at Random (MNAR) [10]. In the case of MCAR, the absence of values occurs without any identifiable pattern or reason. For example, participants may accidentally overlook a question or skip a page in surveys or questionnaires [10]. MAR refers to missing values that are related to other observed variables. For instance, missing values related to gender may occur if male participants are less likely to undergo a depression survey [11]. In the case of MNAR, the missing values are directly related to the values themselves. For example, participants with high alcohol consumption may choose not to answer a question about their drinking habits [10].

The proportion of missing values in a dataset plays a role in selecting appropriate treatment methods. Missing values can be considered trivial if they account for less than 1% of the complete dataset [9]. Percentages between 1% and 5% are still manageable, while percentages between 5% and 15% require more sophisticated approaches for handling missing data. Once the missing data exceeds 15%, it can significantly impact the analysis, necessitating careful treatment measures. Real-world data is often imperfect, unreliable, and noisy. To prepare such data for further computations and analysis, it undergoes preprocessing, which involves cleaning, reducing, transforming, and unifying the data [12]. Preprocessing ensures the quality and fitness of the data for subsequent analysis.

Imputation is the process of replacing missing instances of data with valid values [12]. While one simple approach is to delete vectors containing missing values, this method is often unsuitable [9]. Instead, imputation is a preferred method for treating missing values. In this study, various imputation methods are evaluated, including mean, median, mode, class mean, class median, class mode, random value imputation, k-Nearest Neighbor (k-NN) distance-based median, and several classifier-based imputations such as Support Vector Machine (SVM), Decision tree, and Neural Network (NN).

Classification is a machine learning technique used to predict a group for a given instance based on training data. It enables efficient categorization of new data [13]. Among various classification algorithms, decision trees and Naive Bayes (NB) are commonly used for imputation [14]. Clustering is an unsupervised data mining technique that groups similar data together [15]. In the process of clustering, vectors are grouped to form cohesive clusters containing similar elements. The quality of a cluster is determined by the closeness of elements within the cluster and the distance of cluster elements from non-cluster elements [15]. To assess the identified imputation methods, this study employs two clustering methods: k-means and PAM clustering.

The motivation for this research is primarily driven by the many drawbacks of missing values, including classification inaccuracy, mining inconsistencies, and certain algorithms' inability to process data. One of the imputation methods to address missing values involves imputation based on domain knowledge. However, evaluating such imputations is challenging due to the actual missing values' absence. This study explores different imputation methods to substitute near-real values for missing values and the effect of such imputations on various data mining algorithms. The research employs supervised mining methods to evaluate the proposed imputation methods' impacts. The research contributes by identifying different imputation methods for the missing data value problem, evaluating the performance of these methods, identifying the best technique for imputing missing values in ordinal data, and assessing whether the imputed values for ordinal data can improve data clustering. This research is driven by the following objectives:

- Evaluate and compare various imputation techniques for handling missing ordinal data.
- Assess the impact of imputation techniques on clustering performance and quality using clustering algorithms like k-means and Partitioning Around Medoids (PAM).
- Analyze the effect of imputed values on classification accuracy for various algorithms including k-Nearest Neighbors (kNN), Naive Bayes (NB), and Multilayer Perceptron (MLP).

This study uses a quantitative approach and leverages open-source tools like R-language [16] and WEKA [17] for data analysis. Several benchmark datasets with ordinal values from the UCI machine learning repository [18] are selected. These datasets are then manipulated to create subsets with varying degrees of missing values: 1%, 5%, 10%, 15%, 20%, and 25%. Various imputation methods are accessed to fill in the missing values. The effectiveness of these methods is evaluated by applying different clustering algorithms, such as k-means and Partitioning Around Medoids (PAM), to the completed datasets and comparing the clustering results with the original datasets. This analysis provided insights into the impact of imputation techniques on the clustering performance and allowed for a comprehensive evaluation of the imputation methods' effectiveness. In addition to clustering analysis, classification experiments are conducted to evaluate the effect of imputed values on the accuracy of various classification algorithms, including k-Nearest Neighbors (kNN), Naive Bayes (NB), and Multilayer Perceptron (MLP). By assessing the classification accuracy of the imputed datasets compared to the original datasets, the study provides insights into the impact of different imputation techniques on the performance of classification algorithms in the context of ordinal data. We have made the following contributions:

- We have conducted comprehensive experiments on five datasets, each comprising six subsets containing missing values, by the evaluation of various methods for handling missing values in ordinal datasets and found the most suitable technique for imputing missing instances, specifically in ordinal data.
- We have performed a comprehensive evaluation of clustering performance by measuring the impact of these eleven imputation techniques on clustering performance using four key metrics, including Sum of Squares Within clusters (SSW), cluster diameter, average intra-cluster distance, and average inter-cluster distance.
- We have performed the evaluation of classification performance using various datasets and diverse machine learning algorithms to analyze the impact of imputation methods on classification accuracy.

The rest of the paper is structured as follows: Section 2 delves into the existing research and establishes the context for the study. Section 3 outlines the proposed experimental methodology for data collection and performance assessment. Section 4 performs an in-depth analysis of the results and findings obtained through rigorous experimentation, providing valuable insights and interpretations. The last section summarizes the key findings and their implications, discusses the limitations of the study, and suggests potential avenues for future research.

2. Related work

This section provides a summary of existing research conducted in the field of data analysis related to topics such as ordinal data, missing data, imputation methods, and clustering algorithms. Data quality has become a crucial factor in Data Mining and Business Intelligence, as incomplete and noisy data pose challenges in analysis and can lead to erroneous statistical inferences and decision-making [11,19]. Therefore, it is essential to address missing values before performing any analysis. The summary of the commonly used techniques for treating missing values is presented in Table 1.

Various methods have been developed to handle missing values in numeric data, although their suitability for ordinal datasets may

Table 1
Summary of Related Approaches Missing Data Imputation.

Work	Description
[5]	Joreskog's classification of missing data types
[3]	Factors contributing to missing values in surveys
[8]	Imputation techniques for improving data analysis
[2]	Challenges posed by missing data in analysis
[1]	Impact of missing data on decision-making
[6]	Imputation as a solution for replacing missing data
[7]	Uninformed patterns of absent data in real-world datasets
[18]	UCI Machine Learning Repository for benchmark datasets
[4]	Critique of discarding data vectors with missing values
[16]	R-language as an open-source tool for data analysis
[20]	Artificial Neural Networks for model-based imputation
[21]	Evaluation of k-Nearest Neighbors imputation method
[22]	Missing data imputation using predictive models
[23]	Impact of removing samples with high missing values proportion
[24]	Determining the acceptable threshold for missing values
[25]	Limitations of simple statistical imputation approaches
[26]	Systematic review of data imputation methods in data mining
[27]	Data imputation for missing values and ensuring data integrity
[28]	Regression techniques for model-based imputation
[29]	Deep Learning-based imputation approaches
[30]	Comparison of dynamic imputation techniques

vary [11]. In the context of treating missing values, several commonly used techniques are worth mentioning.

Case Deletion (CD) involves discarding vectors that contain missing data, resulting in a new dataset for further processing. However, this method is not appropriate for datasets with a higher degree of missing values. Even for datasets with smaller quantities of missing data, caution must be exercised to evaluate potential bias introduced by the resultant dataset [9].

Random Value Imputation can be used to replace missing values, thereby creating a completed dataset. While this approach is straightforward, it does not utilize any information from the dataset and may introduce randomness that affects subsequent analysis [31].

Mean Imputation (MI) replaces missing values with the mean value of the corresponding feature or attribute in the complete dataset, as shown in Eq. (1). This type of replacement is also known as complete mean imputation [5]. An alternative approach involves calculating the mean for a given class, similar to the class of the vector containing the missing value [19]. It is important to note that mean imputation may not be suitable for datasets with numerous missing values, as it reduces variance and tends to overestimate the sample size. Furthermore, mean imputation is not appropriate for ordinal data, given that the concept of mean does not apply in an ordinal scale.

$$\dot{x}ij = \sum i : x_{ij} \in C_k \frac{x_{ij}}{n_k} \tag{1}$$

Most Common Imputation (MCI) method replaces missing values with the most frequently occurring value in the dataset [13,32]. This approach assumes that the most common value represents a plausible estimate for the missing data.

Median Imputation replaces missing values with the median value of the corresponding feature in the dataset. Eq. (2) represents complete median imputation [33]. Alternatively, class median imputation involves replacing missing values with the median of the feature within a specific class. The class should correspond to the class variable of the vector containing the missing value.

$$\dot{x}ij = \text{median}\left(xij : x_{ij} \in C_k\right) \tag{2}$$

The aforementioned imputation methods can be further customized to improve their effectiveness. Mean, median, and mode values can be calculated for each class, increasing the chances of approximating the original values. For example, instead of replacing missing values with the mean income of all families, the mean income of only married samples can be utilized. However, it is important to acknowledge that this approach may introduce bias, as all elements within a group will have the same mean value.

k-Nearest Neighbor (k-NN) distance-based median procedure replaces missing values with k instances that are most similar in terms of relevance [34]. The similarity between instances is determined using a distance metric. The procedure involves splitting the dataset into two fragments: DTm, which represents a subset containing at least one vector with a missing feature or instance, and DTc, which contains the remaining vectors without any missing data. For each row in DTm, the instance row is partitioned into observed and missing sections. The distance between the observed section and the rows of DTc is computed, followed by the identification of the k most similar instances using the k-nearest neighbors approach [34].

Several techniques have been proposed for handling missing data in ordinal datasets. Decision trees have demonstrated effectiveness in the classification of ordinal data, as they facilitate the categorization of data into branches or splits [35]. Traditional approaches for handling missing values can lead to biased estimates and may reduce or exaggerate statistical power [14,36]. Deleting missing instances is often chosen as the quickest method for treating missing data and is the default procedure in many statistical data analysis tools. However, this approach may result in the deletion of a significant portion of the data in real-world scenarios.

Neural networks offer a suitable technique for addressing the missing value problem in ordinal data. They represent a classifier-based imputation technique that emulates the physiological functioning of the brain. Neural networks typically consist of an input layer, a hidden layer, and an output layer. Through training algorithms, neural networks effectively solve complex mathematical equations, with the hidden layer adapting during the process [37,38]. Another classifier-based imputation technique for handling missing data is Support Vector Machines (SVM) [39,40].

Classification-based imputation techniques have emerged as viable approaches for estimating and imputing missing values in datasets [41,42]. These techniques leverage various classification methods, such as neural networks, decision trees, and similar procedures, to address the missing value problem. Notably, previous research has predominantly focused on handling numerical or nominal missing data values [43]. In contrast, this study aims to bridge the existing research gap by specifically addressing the treatment of missing values in ordinal data and investigating the impact of these treatment methods on unsupervised learning techniques, particularly clustering.

Applying classification-based imputation techniques entails constructing a projection model that facilitates the estimation of values closely resembling the actual data, ultimately filling in the missing instances [44]. In this process, the vectors with missing features serve as the reaction vector, while the remaining vectors contribute to the construction of the projection model. However, it is essential to recognize the limitations associated with this approach, as highlighted by Shah et al. [4]. Firstly, the estimated values obtained through classification-based imputation often exhibit superior performance compared to the actual values. Secondly, the accuracy of valuing missing values may be compromised when there are no inherent associations between complete vectors and vectors with missing values. Finally, the computational cost associated with building multiple models for predicting missing values is relatively high. These considerations underscore the need for a critical evaluation of the classification-based imputation techniques when addressing missing values in ordinal data. Additionally, works such as [45] provide essential insights into technologies and phases vital for addressing missing data challenges.

3. Experimental design

In this section, the design of our study for the performance of different imputation techniques in handling missing values across varying datasets and missing value scenarios is thoroughly examined. The implementation does not necessitate any special hardware requirements. The software employed in this study includes widely accessible open-source tools, namely R-language [16] and WEKA [17].

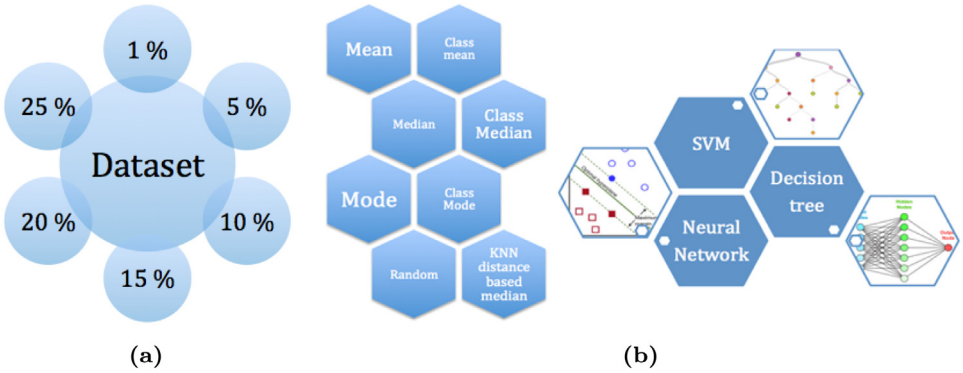

Fig. 1. (a) Data subsets with missing data percentages and (b) imputation methods used for experimentation.

Table 2
Datasets Details.

Dataset	#Instances	#Attributes	Class detail
CAR	1728	5	4 (unacc,acc,good,vgood)
ESL	488	5	9 (1,2,3,4,5,6,7,8,9)
LEV	1000	5	5 (0,1,2,3,4)
ERA	1000	5	9 (1,2,3,4,5,6,7,8,9)
SWD	1000	11	4 (2,3,4,5)

3.1. Dataset description

We have selected five benchmark datasets from the UCI machine learning repository [18] that exhibit the presence of ordinal values, as summarized in Table 2. These datasets serve as the foundation for evaluating diverse imputation methods. Subsequently, the selected datasets undergo processing to generate six additional datasets, each characterized by a distinct proportion of missing values. The experimental outcomes are acquired for subsets derived from the original dataset, comprising missing value proportions of 1%, 5%, 10%, 15%, 20%, and 25%, as visually represented in Fig. 1(a). This systematic approach enables us to thoroughly examine the performance of different imputation techniques in handling missing values across varying datasets and missing value scenarios.

3.2. Problem assessment

A comprehensive evaluation of various data imputation methods is conducted, with a specific focus on their applicability in data mining techniques, particularly clustering and classification. The assessment encompasses a range of imputation techniques, including random imputation, class mean imputation, global mean value imputation, global median value imputation, class mode value imputation, class median value imputation, most common value imputation, and classification-based imputation methods such as Support Vector Machines (SVM), Decision Tree, and Neural Networks (NN), as visually depicted in Fig. 1(b).

3.3. Choice of imputation algorithms

The selection of imputation algorithms was driven by the need to ensure a comprehensive evaluation of methods for handling missing ordinal data within diverse datasets. Each chosen technique serves a specific purpose, allowing us to cover a wide spectrum of imputation strategies.

- **Mean, Median, and Mode Imputation:** These basic techniques were included to provide a fundamental comparison. Mean imputation offers simplicity and efficiency, while median and mode imputation provide robustness against outliers and skewed data, respectively.
- **Class-based Imputation:** Considering class-specific statistics in imputation is crucial for datasets with inherent class structures. Class mean, class median, and class mode imputation ensure that imputed values align with the characteristics of their respective classes, preserving important class-related information.
- **Random Number Imputation:** The randomness introduced by this method allows for variation in the imputed dataset. This is essential in scenarios where uncertainty or variability in missing values is a significant consideration.
- **k-NN Distance-Based Median Imputation:** Utilizing the local neighborhood of a missing value for imputation is essential for datasets with localized patterns. k-NN imputation leverages the proximity of instances to ensure that imputed values reflect the local data structure accurately.
- **SVM and Decision Tree Imputation:** Classification-based imputation techniques are valuable when the relationship between features is complex. SVM and decision tree imputation leverage the power of classification algorithms to predict missing values, capturing intricate data patterns effectively.
- **Neural Network Imputation:** Neural networks, being versatile and powerful models, can approximate complex data patterns. Imputing missing values using a neural network ensures that the imputed dataset aligns well with the original data's underlying patterns.

This diverse selection of imputation techniques enables a thorough evaluation, considering both the inherent characteristics of the missing data and the complexities present within the ordinal datasets.

3.4. Performance assessments

We evaluate the efficacy of different imputation methods in enhancing the learning outcomes for datasets afflicted with missing values. The evaluation centers around various clustering and classification algorithms suitable for ordinal data and compares their performance with that of the original datasets. Specifically, the clustering algorithms employed are k-means and Partitioning Around Medoids (PAM). k-Nearest Neighbors (kNN), Naive Bayes (NB) and Multilayer Perceptron (MLP) are used for classification assessment.

4. Experimental evaluation

In this section, we present a detailed analysis of the results obtained from conducted experimentation. The experimentation encompasses five datasets, each consisting of six subsets with missing data ranging from 1% to 25%. For each subset, eleven imputation techniques are implemented to address the missing values. We discuss the accuracy achieved and the average variance of imputed values compared to the original values for each dataset. We highlight the performance of different imputation techniques, including mean, median, mode, class

Fig. 2. Accuracy and average variance of various imputation techniques for the CAR dataset.

mean, class median, class mode, random number imputation, k-NN distance-based median imputation, SVM, Decision Tree, and Neural Network. Each of these techniques is applied to the datasets to fill in the missing values and obtain completed datasets. The subsequent analysis and evaluation of these imputed datasets allow for a comprehensive assessment of the performance and effectiveness of each imputation technique. Furthermore, we examine the clustering performance of the imputed data using the k-means and PAM algorithms, comparing the clusters formed with the original dataset. This analysis helps us identify the most suitable techniques for addressing the problem of missing values in ordinal data and provide valuable insights for future research and applications.

The choice of clustering and classification algorithms in this study was deliberate to ensure a robust evaluation in the context of ordinal data. For clustering, we utilized k-means and PAM algorithms. k-means offers efficiency, suitable for large datasets, while PAM handles outliers and noise effectively. These algorithms align well with ordinal data, grouping based on ordinal rankings to assess imputed data clusters against the original dataset. In classification evaluation, SVM, Decision Tree, and Neural Network were chosen for their ability to capture nonlinear relationships in data, essential for ordinal data exhibiting such patterns. This selection ensures a meaningful analysis of imputation effects on classification accuracy, aligning with ordinal data's characteristics.

4.1. Imputation performance for ordinal data

This section presents the accuracy achieved and the average variance of imputed values compared to the original values using eleven selected techniques for imputing missing values using each subset of the dataset containing missing values in varying proportions. The accuracy and variance of imputed values highlight the suitability of different imputation methods. The decision tree emerges as a promising technique for imputing missing values in ordinal datasets, demonstrating high accuracy and minimal deviation from the original dataset. The detailed results and findings for each dataset are presented in the subsequent sections.

4.1.1. Imputing missing values in the CAR dataset

Fig. 2 shows the accuracy and average variance across all subsets of the CAR dataset when different imputation methods are applied, represented by different colors. The decision tree method, a way of filling in missing values based on patterns in the data, achieves the best accuracy of 96% in one of the subsets. This high accuracy suggests that the decision tree method works effectively for this type of dataset. On the other hand, the neural network-based method does not perform well, indicating that it might not be the best choice for handling missing values in this dataset.

Two methods that fill in missing data with the most common value, called mode and class mode, show similar results for the CAR dataset,